

Information Security

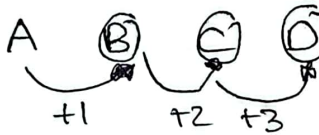
Assignment - 01

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Question 1 Caesar's Cipher

Q1.1 Instructor Name:- SYED MOHAMMAD IRTEZA
Default key:- 3

We will shift each letter 3 letters forward and write that letter. For example if it is A, then it will be D. A B C D E, D will be chosen,



A B C D E
+1 +2 +3

Cipher Text: VBHGPRKDPDGILUWHCD

Q1.2 Cipher Text: ZHORYHSONLVWDQ

Using key value 1:-

~~ALPSZITZOMWYER~~ YGNQXGRCKUVCP

Using key value 2:-

~~BIO~~ XFMPWFQBLJTUBO

Using key value 3:-

~~KRU~~ WELOVEPAKISTAN

~~Using key value 4:-~~

~~BLSEVE~~ Value of 3 is to be used to decrypt ~~et.~~

Plain Text :- We love Pakistan.

↓
cipher
text

In the above question, I simply used a brute force approach and shifted each letter to the letter before it and increased the shifting number at number 3 I got the plain text.

Question 2 Monoalphabetic

Alphabet Frequency

B	4
N	2
C	4
E	5
P	3
Q	4
A	4
A	4
I	2
M	4
F	1
Y	3
H	1
V	1

Ciphertext:- BNCEPQCBEAQBE BIMQFN IYQAEAEYCPMH
Plaintext:- A R E T H O E A T ~~T~~ O A T A M P O N R M S O I T I T S E W P L

Ciphertext:- VMAYCMP
Plaintext:- B P I S E P H

- ① The most common word in english is e ~~and~~ and the most common word in ciphertext is also e, so hence we will use the second most common word that is ~~t~~.
- ② We will substitute the second most common words in cipher text with the most common words in english. replace B with A, C ~~with A~~, A with E, Q with O, A with I.
- ③ Replace more characters and try to derive english words.
I ~~was unable to~~ solve it as there are too many possibilities and cannot do this manually, it is very hard as I don't know which letter is mapped onto which letter hence it is ~~very hard~~ ~~not possible~~ to decrypt this cipher text.

Question 3 Vigenère Cipher

Plaintext = KTWOISTHEHIGHESTPEAKOF
PAKISTAN

KEY = FASTNUES

Note this can also be solved by using alphabets matrix
take the below method is also one of the methods.

Encryption

Plaintext letters	KEY	Addition of their respective number codes and the mod 26	Encrypted Text
K	F	$(10+5) \cdot 1 \cdot 26$	15 — P
T	A	$(19+0) \cdot 1 \cdot 26$	19 — T
W	S	$(22+18) \cdot 1 \cdot 26$	14 — O
O	T	$(14+19) \cdot 1 \cdot 26$	7 — H
I	N	$(8+13) \cdot 1 \cdot 26$	21 — Y
S	U	$(18+20) \cdot 1 \cdot 26$	12 — M
T	C	$(19+2) \cdot 1 \cdot 26$	21 — Y
H	E	$(7+4) \cdot 1 \cdot 26$	11 — L
E	S	$(4+18) \cdot 1 \cdot 26$	22 — W
H	F	$(7+5) \cdot 1 \cdot 26$	12 — M
I	A	$(8+0) \cdot 1 \cdot 26$	8 — I
G	S	$(6+18) \cdot 1 \cdot 26$	24 — Y
H	T	$(7+19) \cdot 1 \cdot 26$	0 — A
E	N	$(4+13) \cdot 1 \cdot 26$	17 — R
S	U	$(18+20) \cdot 1 \cdot 26$	12 — M
T	C	$(19+2) \cdot 1 \cdot 26$	21 — Y
P	E	$(15+4) \cdot 1 \cdot 26$	19 — T
E	S	$(4+18) \cdot 1 \cdot 26$	22 — W
A	F	$(0+5) \cdot 1 \cdot 26$	5 — F
K	A	$(10+0) \cdot 1 \cdot 26$	10 — K
O	S	$(14+18) \cdot 1 \cdot 26$	6 — G
F	T	$(5+19) \cdot 1 \cdot 26$	24 — Y
P	N	$(15+13) \cdot 1 \cdot 26$	2 — C
A	U	$(0+20) \cdot 1 \cdot 26$	20 — U
K	C	$(10+2) \cdot 1 \cdot 26$	12 — M
I	E	$(8+4) \cdot 1 \cdot 26$	12 — M
S	S	$(18+18) \cdot 1 \cdot 26$	10 — K
T	F	$(19+5) \cdot 1 \cdot 26$	24 — Y
A	A	$(0+0) \cdot 1 \cdot 26$	0 — A
N	S	$(13+18) \cdot 1 \cdot 26$	5 — F

Encrypted Text = PTOHYMYLWMIYARMYTWF

YCUMMKYAF

CipherText Letters	KEY	Subtraction of their respective numbers and take mod 26	Decrypted Text
P	F	$(15 - 5) \cdot 1 \cdot 26$	10 — K
T	A	$(19 - 0) \cdot 1 \cdot 26$	19 — T
O	S	$(14 - 18) \cdot 1 \cdot 26$	22 — W
H	T	$(7 - 19) \cdot 1 \cdot 26$	14 — O
Y	N	$(21 - 13) \cdot 1 \cdot 26$	8 — I
M	U	$(12 - 20) \cdot 1 \cdot 26$	18 — S
Y	C	$(21 - 2) \cdot 1 \cdot 26$	19 — T
L	E	$(11 - 4) \cdot 1 \cdot 26$	7 — H
W	S	$(22 - 18) \cdot 1 \cdot 26$	4 — E
M	F	$(12 - 5) \cdot 1 \cdot 26$	7 — H
I	A	$(8 - 0) \cdot 1 \cdot 26$	8 — I
Y	S	$(24 - 18) \cdot 1 \cdot 26$	6 — G
A	T	$(0 - 19) \cdot 1 \cdot 26$	7 — H
R	N	$(17 - 13) \cdot 1 \cdot 26$	4 — E
M	U	$(12 - 20) \cdot 1 \cdot 26$	18 — S
Y	C	$(21 - 2) \cdot 1 \cdot 26$	19 — T
T	E	$(19 - 4) \cdot 1 \cdot 26$	15 — P
W	S	$(22 - 18) \cdot 1 \cdot 26$	4 — E
F	F	$(5 - 5) \cdot 1 \cdot 26$	0 — A
K	A	$(10 - 0) \cdot 1 \cdot 26$	10 — K
G	S	$(6 - 18) \cdot 1 \cdot 26$	14 — O
Y	T	$(24 - 19) \cdot 1 \cdot 26$	5 — F
C	N	$(2 - 13) \cdot 1 \cdot 26$	15 — P
U	U	$(20 - 20) \cdot 1 \cdot 26$	0 — A
M	C	$(12 - 2) \cdot 1 \cdot 26$	10 — K
M	E	$(12 - 4) \cdot 1 \cdot 26$	8 — I
K	S	$(10 - 18) \cdot 1 \cdot 26$	18 — S
Y	F	$(24 - 5) \cdot 1 \cdot 26$	19 — T
A	A	$(0 - 0) \cdot 1 \cdot 26$	0 — A
F	S	$(5 - 18) \cdot 1 \cdot 26$	13 — N

Decrypted Text = KTWOLSTHE HIGHEST PEAK OF PAKISTAN

Question 4. Row Transposition Encryption

Key:- I R T E Z A

Lexicographic

~~Order~~ = 3 4 5 2 6 1

4.1

1 1 1 1 1 1
3 4 5 2 6 1

3	4	5	2	6	1
A	T	T	A	C	K
P	O	S	T	P	O
N	E	D	U	N	T
I	L	T	W	O	A
M	*	*	*	*	*

NOTE!
* are being placed as during decryption, it won't be possible to get the no. of letters in each column as all the rows are not filled.

KOTA* ATUW* APNIMTOEL* TSDT* CPNO*

Cipher Text = ~~TSDT* ATUW* APNIMTOEL* KOTA* CPNO*~~

Decryption

Key = I R T E Z A

No of letters = 6

Lexicographic ~~Order~~ = 3 4 5 2 6 1 Columns ~~Rows~~ = 6

3	4	5	2	6	1
A	T	T	A	C	K
P	O	S	T	P	O
N	E	D	U	N	T
I	L	T	W	O	A
M	*	*	*	*	*

Number of letters in each column

= Total no. of letters

Number of letters in key

= $\frac{30}{6} = 5$ letters

Plaintext: ATTACKPOSTPONED UNTIL TWO AM

steps ① Write the plaintext in the matrix, the columns of the matrix should be the number of letters in the key.

② Arrange the key in ~~lexicographic~~ order and give them numbers, then ~~see the original key's number and then write the contents of that specific columns first and then~~ ~~as received go on.~~ number wise.

③ For Decryption calculate the number of ~~column~~ in each ^{column} by Total no. of letters in plain divided by no. of letters in key. Then take many number of letters from the cipher text & start placing them in the matrix. Again if the key is IRTEZA the **I** letters represents **6** column ^{according to order} so the number of letters should be placed in the **6** column. At last read the whole message from left to right.

4.2 Key = PAKISTAN Order = 61437825

Encryption

	6	1	3	1	2	1	
	4	7	8				
	6	4	4	3	7	8	2
	5						
S	A	V	E	Y	O	U	R
S	E	L	F	W	E	A	R
E	D	I	S	C	O	V	E
R	E	D	*	*	*	*	*

CipherText = AEDEUAV*EFS*VLIDRRE*SSERYWC*OEO*
~~CEO*SSERETS*VLIDUAV*RRR~~
~~AEDUAV*WC*~~

Decryption

Number of letters in each ^{column} row = $32/8 = 4$

	6	4	4	3	7	8	2
	5						
S	A	V	E	Y	O	U	R
S	E	L	F	W	E	A	R
E	D	I	S	C	O	V	E
R	E	D	*	*	*	*	*

Breakdown of cipher text

= OEO* SSER EFS* VLID UAV* RRE* AEDE YWC*
 6 4 3 7 8 2 5

Give them numbering according to the key = PAKISTAN

Plaintext = SAVE YOUR SELF WE ARE DISCOVERED,

The only ~~challenge~~ challenges faced was the placing of * to maintain the position of the letters in columns.

		A			E		I		O		E		I		O	
H	L		H	R	E	R	I	L	C	I	D	N	H	N	P	
	E		D		D		S		A		I		E			

	E		A		E		D		O		A		D		Z	
H	R	P	R	O	L	H	R	S	L	E	L	E	C	I		
I		N		I		A		E		D		L		I		

- Steps ① Make a grid of Number of letters x Depth size
- ② Outline all the places where the cipher's text letters will be placed.
- ③ First fill all the letters in the first row in the outlined spaces
Once filled, then go to 2nd row and so on untill all letters are filled.
- ④ Read the ^{decrypted.} message

DECRYPTION

Plain Text = THE LAHORE FORT IS LOCATED IN THE
NORTHERN PART OF LAHORES OLD WALLED CITY



Lesson 6: Rail Fence

- Number of letters = 23

- Depth level = 2

W		W		L		H		V		A		U		Z		Z		O		D		Y
	E		I		L		A		E		Q		I		O		M		N		A	

Decrypted = We will have a quiz on monday
Text

